133384: leucogabbro, Thompsons Bore

(Trillbar Formation, Capricorn Orogen)

Location and sampling

Robinson Range (SG 50-7), Moorarie (2446)
MGA Zone 50, 573839E 7173032N

Sampled on 6 June 2004

This sample was collected from one of several low boulders to the east of a creek on Mount Gould Station, about 6.3 km north-northeast of Mount Seabrook, 3.6 km northwest of Trillbar homestead, and 3.6 km east-northeast of Thompsons Bore.

Geological context

The unit sampled is the Trillbar Formation, which consists of metamorphosed layered gabbro, microgabbro, leucogabbro and melanogabbro (Myers, 1989; Occhipinti and Myers, 1999). The Trillbar Formation is entirely fault-bounded and its original relationships with surrounding units are unknown (Occhipinti et al., 2017). The Trillbar Formation was previously considered a part of the 2014–1920 Ma Bryah Group of the Bryah Basin (Johnson and Thorne, 2022). However, the age of c. 2069 Ma for this sample (detailed below) suggests that the Trillbar Formation is significantly older than the Bryah Group. Geochemical data for mafic and ultramafic rocks suggest a mid-ocean ridge (MORB) signature that the Trillbar Formation formed at a mid-ocean ridge, whereas the Bryah Group is thought to have formed in an intracontinental rift setting (Olierook et al., 2018).

Petrographic description

The sample is a foliated, leucocratic metagabbro, consisting of medium-grained phenocrysts of clinopyroxene (altered to actinolite) within a fine-grained matrix of plagioclase and clinopyroxene with minor sericite alteration. A detailed petrographic description is not available for this sample.

Zircon morphology

Zircons from this sample are anhedral, and occur as single crystals or in aggregates. The zircons are up to 150 µm long, and equant to elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, most crystals display concentric zoning. A CL image of all zircons is shown in Figure 1.

Analytical details

This sample was analysed on 20 February 2005, using SHRIMP-B. Zircons were analysed in plugs cut from polished thin sections, following procedures described by Olierook et al (2018). Eleven analyses of the CZ3 zircon standard were obtained during the session, of which 10 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.65% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.65% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-one analyses were obtained from six zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 2).

Interpretation

The analyses are concordant to moderately discordant (Fig. 2). One analysis is >5% discordant and one indicates high common Pb ($f204$ >1%). The dates obtained from these two analyses (Group D, Table 1) are unreliable, and considered not to be geologically significant. The remaining 19 analyses form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 19 analyses of six zircons (Table 1), which yield a weighted $^{207}Pb^*/^{206}Pb^*$ date of 2069 ± 9 Ma (MSWD = 1.3).

The date of 2069 ± 9 Ma for the 19 analyses in Group I is interpreted as the magmatic crystallization age of the leucogabbro. The high and variable Th/U ratios of the zircons (0.27 – 3.19, median 1.74) are typical of primary zircons in mafic intrusions and evidence that they are not inherited (e.g. Lanyon et al., 1993).

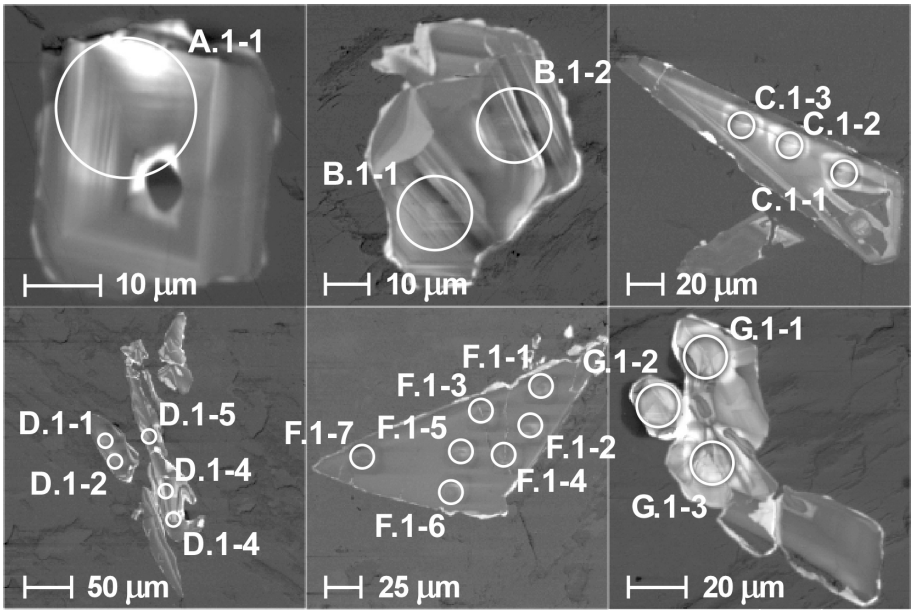

Figure 1. Cathodoluminescence images of zircons from sample 133384: leucogabbro, Thompsons Bore. Numbered circles indicate the approximate locations of analysis sites

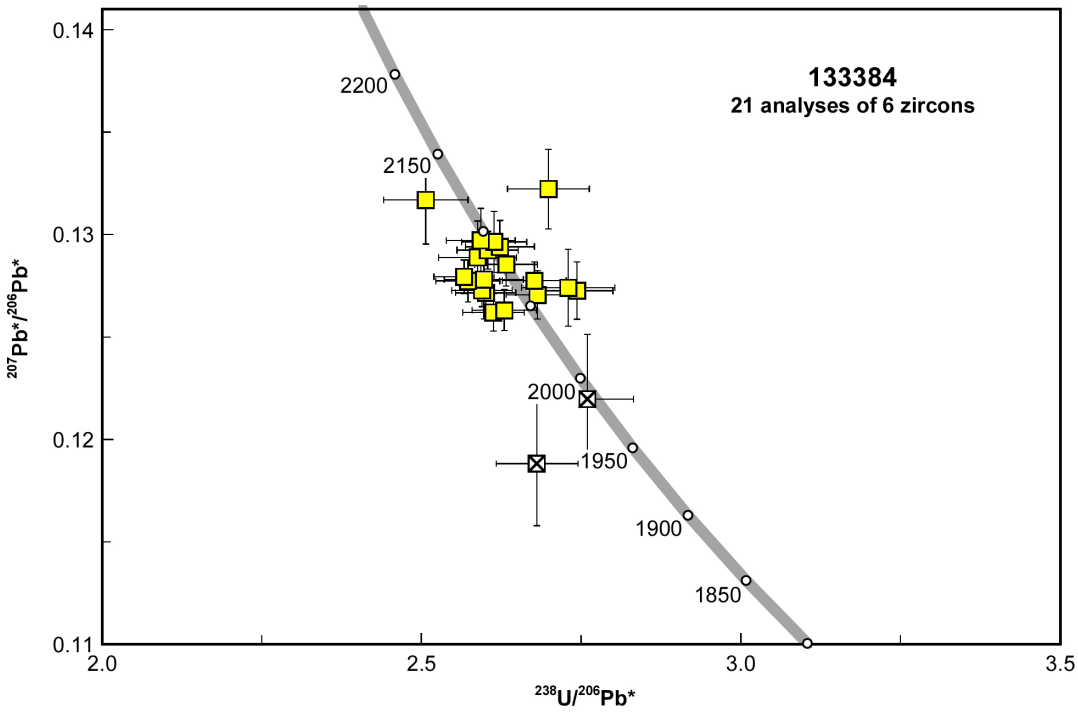

Figure 2. U–Pb analytical data for sample 133384: leucogabbro, Thompsons Bore. Yellow squares indicate Group I (magmatic zircons); crossed squares indicate Group D (discordance >5% or high common Pb)

Table 1. Ion microprobe analytical results for zircons from sample 133384: leucogabbro, Thompsons Bore

Group ID	Mount no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	BR0510	F.1-4	371	1045	2.92	0.083	2.611	0.048	0.12692	0.00086	2.613	0.048	0.12619	0.00091	2089	33	2046	13	-2.1
I	BR0510	F.1-6	392	1209	3.19	0.057	2.629	0.050	0.12680	0.00098	2.630	0.050	0.12630	0.00100	2077	34	2047	14	-1.5
I	BR0510	D.1-1	376	859	2.36	0.391	2.671	0.048	0.13050	0.00077	2.682	0.049	0.12705	0.00118	2043	32	2058	16	0.7
I	BR0510	F.1-1	444	941	2.19	0.070	2.600	0.047	0.12777	0.00079	2.601	0.047	0.12715	0.00084	2097	33	2059	12	-1.8
I	BR0510	D.1-4	178	291	1.69	0.142	2.740	0.056	0.12851	0.00131	2.744	0.056	0.12726	0.00140	2003	35	2060	19	2.8
I	BR0510	B.1-1	482	128	0.27	0.061	2.593	0.047	0.12781	0.00077	2.595	0.047	0.12727	0.00079	2101	33	2061	11	-2.0
I	BR0510	C.1-1	109	172	1.64	0.220	2.724	0.073	0.12933	0.00167	2.730	0.073	0.12740	0.00187	2012	46	2062	26	2.5
I	BR0510	D.1-5	260	520	2.07	-0.078	2.575	0.050	0.12704	0.00102	2.573	0.050	0.12773	0.00103	2116	35	2067	14	-2.4
I	BR0510	F.1-5	470	1307	2.87	0.000	2.677	0.050	0.12774	0.00092	2.677	0.050	0.12774	0.00092	2046	33	2067	13	1.0
I	BR0510	A.1-1	108	236	2.26	0.035	2.597	0.062	0.12811	0.00175	2.598	0.062	0.12780	0.00193	2099	43	2068	27	-1.5
I	BR0510	F.1-2	408	1105	2.80	0.033	2.566	0.047	0.12822	0.00081	2.567	0.047	0.12793	0.00083	2121	33	2070	11	-2.5
I	BR0510	F.1-7	500	1102	2.28	0.108	2.630	0.049	0.12948	0.00102	2.633	0.049	0.12854	0.00106	2075	33	2078	14	0.1
I	BR0510	G.1-1	82	87	1.09	-0.080	2.590	0.061	0.12818	0.00178	2.588	0.061	0.12888	0.00178	2106	42	2083	24	-1.1
I	BR0510	F.1-3	375	780	2.15	0.075	2.602	0.048	0.12989	0.00087	2.604	0.048	0.12923	0.00092	2095	33	2087	12	-0.4
I	BR0510	D.1-2	195	329	1.74	0.075	2.621	0.054	0.13006	0.00127	2.623	0.054	0.12940	0.00128	2082	37	2090	17	0.4
I	BR0510	B.1-2	268	210	0.81	-0.102	2.617	0.051	0.12874	0.00148	2.614	0.051	0.12964	0.00148	2088	35	2093	20	0.2
I	BR0510	D.1-3	145	201	1.43	-0.235	2.599	0.054	0.12765	0.00126	2.593	0.054	0.12971	0.00156	2103	37	2094	21	-0.4
I	BR0510	C.1-2	65	90	1.42	-0.408	2.518	0.067	0.12810	0.00211	2.507	0.066	0.13169	0.00216	2164	49	2121	29	-2.0
I	BR0510	C.1-3	104	155	1.55	0.235	2.692	0.064	0.13428	0.00182	2.699	0.064	0.13222	0.00194	2032	41	2128	26	4.5
D	BR0510	G.1-3	63	57	0.94	0.849	2.658	0.063	0.12638	0.00184	2.681	0.064	0.11882	0.00304	2043	41	1938	46	-5.4
D	BR0510	G.1-2	54	70	1.33	1.015	2.732	0.071	0.13098	0.00223	2.760	0.072	0.12197	0.00316	1993	44	1985	46	-0.4

References

Johnson, SP and Thorne, AM (partial report), Bryah Group (P_-BY-xsf-mw): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 22 March 2022 <www.dmp.wa.gov.au/ens>.

Lanyon, R, Black, LP and Seitz, H 1993, U-Pb zircon dating of mafic dykes and its application to the Proterozoic geological history of the Vestfold Hills, East Antarctica: Contributions to Mineralogy and Petrology, v. 115, p. 184–203.

Myers, JS 1989, Thrust sheets on the southern foreland of the Capricorn Orogen, Robinson Range, Western Australia: Professional Papers, Geological Survey of Western Australia, Report 26, p. 127–130.

Occhipinti, SA, Hocking, RM, Lindsay M, Aitken, ARA, Copp, I, Jones, J, Sheppard, S, Pirajno, F and Metelka, V 2017, Paleoproterozoic basin development on the northern Yilgarn Craton, Western Australia: Precambrian Research, v. 300, p. 121–140.

Occhipinti, SA and Myers, JS 1999, Geology of the Moorarie 1:100 000 sheet: Geological Survey of Western Australia, 1:100 000 Geological Series Explanatory Notes, 20p.

Olierook, HKH, Sheppard, S, Johnson, SP, Occhipinti, SA, Reddy, SM, Clark, C, Fletcher, IR, Rasmussen, B, Zi, J-W, Pirajno, F, LaFlamme, C, Do, T, Ware, B, Blandthorn, E, Lindsay, M, Lu, Y, Crossley, RJ and Erickson, TM 2018, Extensional episodes in the Paleoproterozoic Capricorn Orogen, Western Australia, revealed by petrogenesis and geochronology of mafic–ultramafic rocks: Precambrian Research, v. 306, p. 22–40.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Recommended reference for this publication

Fielding, IOH, Wingate, MTD, Fletcher, IR, Rasmussen, B, Zi, J-W and Sheppard, S 2022, 133384: leucogabbro, Thompsons Bore; Geochronology Record 1860: Geological Survey of Western Australia, 4p.

Data obtained: 20 February 2005

Data released: 14 April 2022